

Dual-Camera Active Acquisition for Automated Small-Object Dataset Construction

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Motivation: The Small-Object Detection Bottleneck

Problem:

- Small objects in wide-angle views occupy $\sim 15 \times 15$ pixels
- Insufficient detail for classification
- Distractors indistinguishable from targets

Existing Approaches:

- **Digital zoom:** Magnifies ambiguity, no new information
- **Super-resolution:** Generative, risk of hallucination, 300+ ms latency
- **Single high-res camera:** Prohibitive GPU memory for 4K/8K real-time

Our Solution: Replace digital zoom and SR with *optical zoom* via dual-camera active acquisition

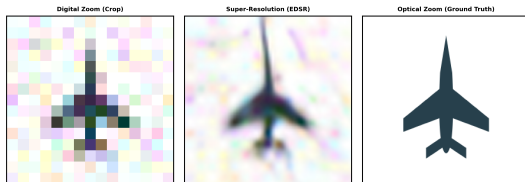


Figure: *

Wide-angle crop vs super-resolution vs optical zoom

Hardware Configuration:

- 2× FLIR M300C cameras
- Baseline separation: 0.5 m
- Parallax at 100 m: 0.29° (negligible)
- Power: 12.8 V 50 Ah LiFePO4 battery
- Compute: NVIDIA Jetson Orin AGX
- JetPack 4.7, CUDA 12.6
- PCIe Ethernet with IEEE 802.3at PoE

Camera Roles:

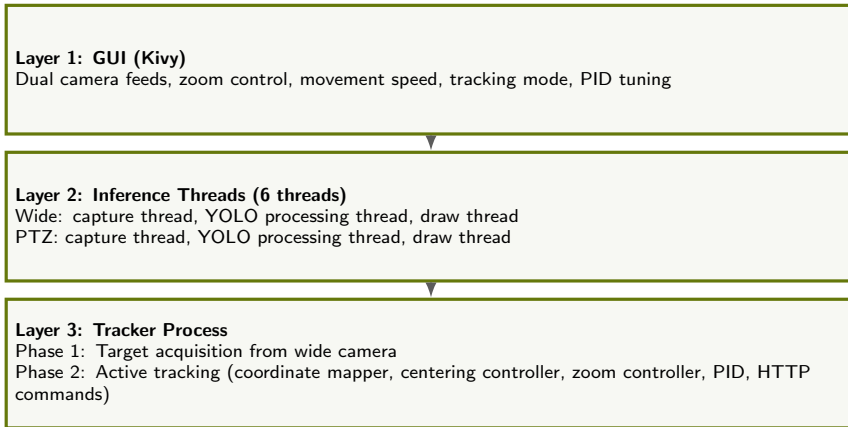
- Wide-angle (left): Persistent surveillance
- PTZ (right): Optical verification on demand



Figure: *

Two FLIR M300C cameras mounted on 0.9 m stand

Software Architecture: Three-Layer Design



Key Feature: Automatic fallback to wide-camera tracking if PTZ loses detection; resumes PTZ control on reacquisition

Bearing Transfer (wide to PTZ):

Bearing from wide detection (u, v) :

$$\hat{d}_w = \frac{K_w^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}}{\left\| K_w^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \right\|}$$

Transform to PTZ base frame:

$$\hat{d}_p = R_{pw} \hat{d}_w$$

PTZ pan/tilt commands:

$$\theta = \text{atan2}(\hat{d}_{p,y}, \hat{d}_{p,x})$$

$$\phi = \text{atan2} \left(\hat{d}_{p,z}, \sqrt{\hat{d}_{p,x}^2 + \hat{d}_{p,y}^2} \right)$$

Zoom Selection by Angular Size:

Angular height from wide detection:

$$\alpha \approx \frac{h^w}{f_w}$$

Desired PTZ pixel height:

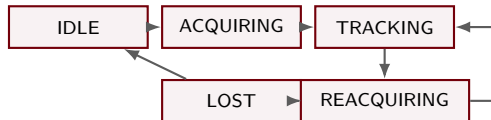
$$h_{\text{des}}^p \approx \frac{h^w}{f_w} f_p(z)$$

Select zoom z to match $f_p(z)$

Key Property:

- Direction-only transfer sufficient for distant targets (no range estimation needed)

Five-State Machine:



PID Controller (30 Hz):

- $K_p = 1.5$, $K_i = 0.05$, $K_d = 1.0$
- Inverse-square speed scaling
- 2% centering deadzone
- Max PTZ speed: 30 deg/s

Predictive Bearing Compensation:

Constant-velocity model with EMA smoothing:

$$\mathbf{v}_t = \alpha \mathbf{v}_{t-1} + (1 - \alpha) \mathbf{v}_{\text{inst}}$$

Predicted bearing ($\alpha = 0.3$):

$$\hat{d}_w(t + \Delta) = d_w(t) + \mathbf{v}_t \cdot \Delta$$

Triggering Policy:

- Largest detected target with confidence > 0.40
- Sequential processing (one PTZ resource)
- 10 s acquisition timeout

Label Transfer Pipeline:

- 1 Wide camera detects ambiguous targets
- 2 PTZ slews and zooms to verify optically
- 3 PTZ detector confirms target class (> 0.35 confidence)
- 4 Map PTZ bounding box back to wide view

Direction-Only Projection:

$$\hat{d}_k^w = R_{wp} \hat{d}_k^p$$

$$\begin{bmatrix} u_k^w \\ v_k^w \\ 1 \end{bmatrix} \propto K_w \hat{d}_k^w$$

Quality Gates:

- PTZ confidence > 0.35 (track threshold)
- Class agreement (wide vs PTZ)
- Motion blur check (Laplacian variance)
- Exposure saturation check
- Frame-boundary rejection

Dataset Schema:

- Wide frame or short clip
- Transferred bounding box
- PTZ telemetry (θ, ϕ, z)
- Detector scores (wide + PTZ)
- Quality metrics

Parallel GPU/DLA Inference:

Camera	Wide (GPU)	PTZ (DLA)
Model	YOLOv11s	YOLOv11n
Resolution	1024×1024	640×640
Inference (ms)	27.9	46.5
Throughput (FPS)	35.9	21.5
Detection rate	100%	99%
Avg confidence	0.553	0.529

Aggregate throughput: 57.4 FPS

True parallel execution (no GPU time-slicing)

Codec Impact on Latency:

- **H.264:** 2–4 frames decode buffering (66–133 ms latency)
- **MJPEG:** Independent frame encoding (15 ms decode latency)
- **Savings:** 60–130 ms reduction

Key Insight:

- GPU 2.2–2.8× faster than DLA per model
- DLA eliminates CUDA context switch overhead
- Enables true concurrent inference

Results: Drone Tracking Demonstration

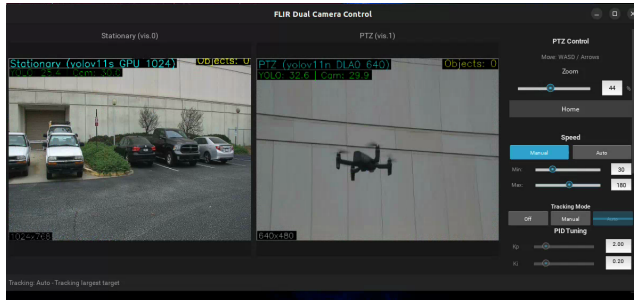


Figure: *

Live GUI during 42-second drone tracking test

Test Parameters:

- Duration: 42 seconds
- Total frames: ~42,000 (both cameras)
- Target: Small commercial drone
- Zero reacquisitions (continuous lock)

Performance:

- Wide detection: 100% rate
- PTZ detection: 99% rate
- Smooth state transitions: IDLE → ACQUIRING → TRACKING
- PID maintained stable centering throughout

Results: Resolution Comparison

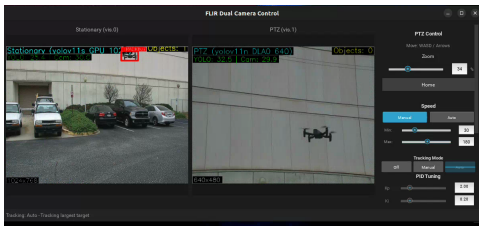


Figure: *

PTZ zoomed view: $\sim 120 \times 90$ px

PTZ View Benefits:

- $6\times$ linear resolution gain
- Moves target into high-accuracy regime (99.1% mAP50)
- Optical ground truth (no hallucination)

Resolution Analysis:

Wide-Angle View

- Target size: $\sim 20 \times 15$ px
- Insufficient for classification
- Distractors indistinguishable

PTZ Zoom View (35–45%)

- Target size: $\sim 120 \times 90$ px
- $6\times$ linear increase
- Clear structural detail

Key Advantage: Optical zoom provides *true* high-resolution data

System Integration: Dual-Instance Pipeline

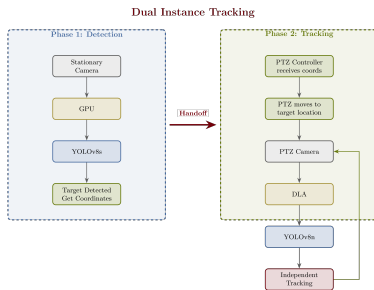


Figure: *

Dual-instance tracking pipeline: Wide detector on GPU, PTZ detector on DLA

Pipeline Features:

- Concurrent detection streams (no resource contention)
- Automatic state machine management (5 states)
- Real-time label transfer with quality gates
- Battery-powered for field deployment

Contributions:

- Dual-camera system for automated small-object dataset construction
- Predictive slew-to-classification control
- Parallel GPU/DLA inference (57.4 FPS)
- Validated on 42-second drone tracking

Key Results:

- $6\times$ resolution gain via optical zoom
- 100% wide, 99% PTZ detection rate
- MJPEG reduces latency by 60–130 ms
- Zero reacquisitions during tracking

Current Limitations:

- Single PTZ (no multi-target zoom)
- Potential loss during slew for fast targets
- Small-parallax assumption for label transfer

Future Work:

- Label quality evaluation vs manual annotation
- Custom YOLO models trained on FGVC aircraft data
- Benchmark COCO vs domain-specific weights
- Multi-PTZ scheduling for higher throughput
- Field deployment with extended battery testing

Questions?

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